

- 2 (a) Current passing A = $0.50 / 5.0 = 0.10$ A
Current passing B = $0.50 / 2.5 = 0.20$ A

p.d. across $R_1 = 5.0$ V

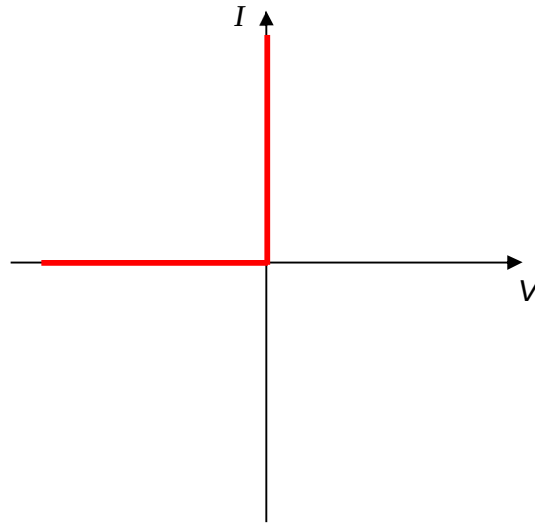
p.d. across $R_2 = 2.5$ V

$$R_1 = 5.0 / 0.30 = 16.7 \, \Omega$$

$$R_2 = 2.5 / 0.20 = 12.5 \, \Omega$$

- (b) $R_3 = 17 \, \Omega$

- (c)



3 (a) The (mutual) electric force acting between two point charges is proportional to product